

Network Configuration & Security | Cisco Packet Tracer

Project Overview

This project shows you how to design, implement and set up the security of a segmented enterprise network using Cisco Packet Tracer.

It was configured with VLANs, IPv4 subnetting, 802.1Q trunking, router-on-a-stick inter-VLAN routing, and extended Access Control Lists (ACLs).

The main goal of the security was to keep different network segments separate, and to deny access to the sensitive network segments (Admin and IT VLANs) from the Guest VLAN, while still providing the necessary network connectivity.

Objectives

Design an enterprise network with segments.

- Design a segmented enterprise network.
- Configure multiple VLANs.
- Implement IPv4 subnetting and IP addressing.
- Configure 802.1Q trunk links.
- Implement router-on-a-stick inter-VLAN routing.
- Configure extended ACLs for network access control.
- Restrict Guest VLAN access to sensitive network segments.
- Test connectivity and security policies.
- Troubleshoot VLAN, routing, trunking, and ACL issues.

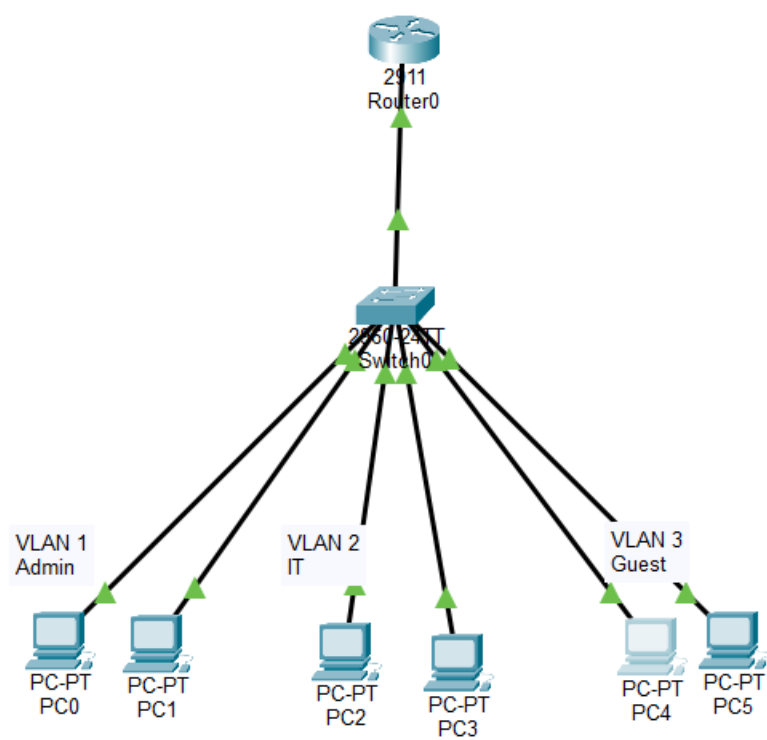
Network Architecture

The network consists of separate VLANs representing different departments and user groups.

VLAN	Department	Network	Default Gateway

10	Admin	192.168.10.0/24	192.168.10.1
20	IT	192.168.20.0/24	192.168.20.1
30	Guest	192.168.30.0/24	192.168.30.1

Network Design



Access Policy

Source	Destination	Access
Admin	IT	Allowed
Admin	Guest	Allowed
IT	Admin	Allowed

IT	Guest	Allowed
Guest	Internet/Permitted Services	Allowed
Guest	Admin	✗ Denied
Guest	IT	✗ Denied

Technologies Used

- Cisco Packet Tracer
- Cisco IOS
- VLANs
- IPv4
- Subnetting
- 802.1Q Trunking
- Router-on-a-Stick
- Inter-VLAN Routing
- Extended ACLs
- Ping Testing
- Network Troubleshooting

Testing

The implementation was tested using Cisco IOS commands and end-device connectivity tests.

Examples:

show vlan brief

show interfaces trunk

show ip interface brief

show access-lists

Connectivity was tested using:

ping 192.168.10.1

ping 192.168.20.1

ping 192.168.30.1

Troubleshooting

When implementing, a number of typical networking configuration problems were taken into account:

- Incorrect VLAN assignment
- Missing VLAN configuration
- Trunk configuration errors
- Incorrect subnet masks
- Incorrect default gateways
- The configuration of the subinterface on the router
- Incorrect ACL placement

Addressing errors that involve the incorrect source/destination addresses: Connectivity errors due to configuration errors The troubleshooting included checking VLAN membership, trunk status, IP addressing, routing configuration, and ACL counters.

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Skills Demonstrated

This project demonstrates practical knowledge of:

- Network segmentation
- VLAN configuration
- IPv4 subnetting
- Cisco IOS
- Inter-VLAN routing
- 802.1Q trunking
- Access Control Lists
- Network security
- Connectivity testing
- Network troubleshooting
- Cisco Packet Tracer

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Project Purpose

This project was developed as a practical networking and cybersecurity exercise to demonstrate how network segmentation and access-control policies can be implemented using Cisco networking technologies.